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CI Practical 4
Problem Statement: Implement Union, Intersection, Complement and Difference operations on fuzzy sets. Also create fuzzy relations by Cartesian product of any two fuzzy sets and perform max-min composition on any two fuzzy relations.

Code:

```
def fuzzy_union(A, B):
    return {x: max(A.get(x, 0), B.get(x, 0)) for x in set(A) | set(B)}

def fuzzy_intersection(A, B):
    return {x: min(A.get(x, 0), B.get(x, 0)) for x in set(A) | set(B)}

def fuzzy_complement(A):
    return {x: 1 - A[x] for x in A}

def fuzzy_difference(A, B):
    return {x: min(A.get(x, 0), 1 - B.get(x, 0)) for x in set(A) | set(B)}

def cartesian_product(A, B):
    return {(x, y): min(A[x], B[y]) for x in A for y in B}

def max_min_composition(R, S):
    T = {}
    for (x, y1) in R:
        for (y2, z) in S:
            if y1 == y2:
                T[(x, z)] = max(T.get((x, z), 0), min(R[(x, y1)], S[(y2, z)]))
    return T

A = {'a':0.5, 'b':0.7, 'c':0.2}
B = {'b':0.6, 'c':0.8, 'd':0.4}
print("A = ",A)
print("B = ",B)

R = cartesian_product(A,B)
print("Cartesian Product (AxB):")
for i in R:
    print(i,R[i])
print("\n")

print("Fuzzy Union: ", fuzzy_union(A,B))
print("Fuzzy Difference: ", fuzzy_difference(A,B))
print("Fuzzy Complement: ", fuzzy_complement(A))
print("Fuzzy Intersection: ", fuzzy_intersection(A,B))
```

```
print("\n")

C = {'x':0.3 ,'y':0.9 ,'z':0.5}
print("C = ",C)
print("Max-Min composition on Two Fuzzy Relations:")
S = cartesian_product(B,C)
max_min = max_min_composition(R,S)
for i in max_min:
    print(i,max_min[i])
```

Output:

```
vboxuser@Ubuntu-24:~/Desktop/CI4$ python3 CI4.py
A =  {'a': 0.5, 'b': 0.7, 'c': 0.2}
B =  {'b': 0.6, 'c': 0.8, 'd': 0.4}
Cartesian Product (AxB):
('a', 'b') 0.5
('a', 'c') 0.5
('a', 'd') 0.4
('b', 'b') 0.6
('b', 'c') 0.7
('b', 'd') 0.4
('c', 'b') 0.2
('c', 'c') 0.2
('c', 'd') 0.2


Fuzzy Union:  {'c': 0.8, 'b': 0.7, 'd': 0.4, 'a': 0.5}
Fuzzy Difference:  {'c': 0.19999999999999996, 'b': 0.4, 'd': 0, 'a': 0.5}
Fuzzy Complement:  {'a': 0.5, 'b': 0.30000000000000004, 'c': 0.8}
Fuzzy Intersection:  {'c': 0.2, 'b': 0.6, 'd': 0, 'a': 0}
```

```
C =  {'x': 0.3, 'y': 0.9, 'z': 0.5}
Max-Min composition on Two Fuzzy Relations:
('a', 'x') 0.3
('a', 'y') 0.5
('a', 'z') 0.5
('b', 'x') 0.3
('b', 'y') 0.7
('b', 'z') 0.5
('c', 'x') 0.2
('c', 'y') 0.2
('c', 'z') 0.2
vboxuser@Ubuntu-24:~/Desktop/CI4$
```